

SWITCH-CASE STATEMENT IN C

Simple Notes + Practice Programs

1. What is a switch-case statement?

The **switch-case** statement is a selection statement in C. It is used when we want to choose one option from several possible options based on the value of an expression.

For example, if the user enters 1, we can perform addition; if the user enters 2, subtraction; and so on.

Important: switch is generally convenient when the choices are fixed values such as 1, 2, 3 or characters such as '+', '-', '*', '/'.

2. Syntax

```
switch(expression)
{
    case value1:
        // statements
        break;

    case value2:
        // statements
        break;

    case value3:
        // statements
        break;

    default:
        // statements
}
```

3. How switch-case works

1. The expression inside **switch()** is evaluated.
2. Its value is compared with each **case** value.
3. When a matching case is found, its statements are executed.
4. The **break** statement exits the switch block.
5. If no case matches, the **default** block is executed.

4. Important keywords

Keyword	Purpose
switch	Starts the switch-case selection block.
case	Defines one possible value.
break	Stops execution of the switch block.
default	Runs when no case matches.

5. Why do we use break?

The **break** statement prevents execution from continuing into the next case. Without break, once a matching case is found, the statements of the following cases may also execute. This is called **fall-through**.

```
switch(choice)
{
    case 1:
        printf("One");
        break;

    case 2:
        printf("Two");
        break;

    default:
        printf("Invalid choice");
}
```

6. Role of default

The **default** case is optional. It is useful for handling an input that does not match any of the given cases.

PROGRAM 1: Calculator using switch-case

This program takes two numbers and an operator from the user. The operator is matched with a case.

```
#include <stdio.h>

int main()
{
    float a, b;
    char op;

    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);

    printf("Enter operation (+, -, *, /): ");
    scanf(" %c", &op);

    switch(op)
    {
        case '+':
            printf("Result = %.2f", a + b);
            break;

        case '-':
            printf("Result = %.2f", a - b);
            break;

        case '*':
            printf("Result = %.2f", a * b);
            break;

        case '/':
            printf("Result = %.2f", a / b);
            break;

        default:
            printf("Invalid operation");
    }

    return 0;
}
```

Practice: Modify the program to add the modulus operator (%) for integer inputs.

PROGRAM 2: Display day of the week

The user enters a number from 1 to 7, and the corresponding day is displayed.

```
#include <stdio.h>

int main()
{
    int day;

    printf("Enter day number (1-7): ");
    scanf("%d", &day);

    switch(day)
    {
        case 1:
            printf("Monday");
            break;

        case 2:
            printf("Tuesday");
            break;
    }
}
```

```
    case 3:
        printf("Wednesday");
        break;

    case 4:
        printf("Thursday");
        break;

    case 5:
        printf("Friday");
        break;

    case 6:
        printf("Saturday");
        break;

    case 7:
        printf("Sunday");
        break;

    default:
        printf("Invalid day number");
}

return 0;
}
```

PROGRAM 3: Vowel or not using switch-case

The user enters a character. The switch statement checks whether it is one of the five vowels.

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    switch(ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            printf("Vowel");
            break;

        default:
            printf("Not a vowel");
    }

    return 0;
}
```

Notice: Multiple case labels can share the same statements. This is useful when different values require the same output.

PROGRAM 4: Simple menu-driven program

This program displays a menu and performs an operation based on the user's choice.

```
#include <stdio.h>

int main()
{
    int choice, n;

    printf("1. Check even/odd\n");
    printf("2. Find square\n");
    printf("3. Find cube\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a number: ");
    scanf("%d", &n);

    switch(choice)
    {
        case 1:
            if(n % 2 == 0)
                printf("Even");
            else
                printf("Odd");
            break;

        case 2:
            printf("Square = %d", n * n);
            break;
    }
}
```

```

        case 3:
            printf("Cube = %d", n * n * n);
            break;

        default:
            printf("Invalid choice");
    }

    return 0;
}

```

Note: The menu demonstrates switch-case. The if-else inside case 1 is only for the even/odd check.

Practice Questions

1. Create a menu to calculate the area of a circle, rectangle, or square using switch-case.
2. Create a program that accepts a month number (1-12) and displays the month name.
3. Create a program that accepts a character and displays whether it is a vowel using switch-case.
4. Create a calculator that accepts two integers and a choice: 1-Add, 2-Subtract, 3-Multiply, 4-Divide.
5. Write a program to display the number of days in a month using switch-case.

Quick Revision

- **switch(expression)** selects among multiple choices.
- Each option is written using **case**.
- Use **break** to stop fall-through.
- **default** handles unmatched values.
- Multiple cases can share the same statements.
- Case values must be constant integral/character values.